

YUE ZHANG

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► RESEARCH INTERESTS

Research interests focus on edge intelligence and resource orchestration for heterogeneous wireless networks, particularly UAV/HAP-assisted and integrated networks. Current interests include predictive multi-timescale network-service co-design, uncertainty-aware resource orchestration, and digital-twin-assisted edge intelligence.

► ACADEMIC APPOINTMENT

Qufu Normal University <i>Lecturer, School of Computer Science</i>	2026.06 - Present <i>Rizhao, China</i>
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► EDUCATION

Dalian Maritime University <i>Ph.D. in Information and Communication Engineering</i> - Supervisors: Prof. Zhenyu Na and Prof. Bin Lin - Dissertation: Research on Coordinated Allocation of Multidimensional Resources for Multi-Layer Aerial Internet of Things	2021.09 - 2025.12 <i>Dalian, China</i>
Dalian University <i>M.S. in Computer Technology</i>	2018.09 - 2021.06 <i>Dalian, China</i>
Binzhou University <i>B.S. in Computer Science and Technology</i>	2012.09 - 2017.06 <i>Binzhou, China</i>

► PUBLICATIONS

First-Author Journal Articles

- Y. Zhang**, Z. Na, L. Jiang, A. Nallanathan, and X. Liu. "Dwell-time-constrained joint task offloading and resource allocation for multi-layer aerial vehicular edge computing networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 27, no. 7, pp. 7805-7819, 2026.
- Y. Zhang**, Z. Na, B. Lin, Y. Lin, and A. Nallanathan. "Energy consumption minimization for integrated sensing, communication, computing and caching in multi-layer aerial Internet of Things," *IEEE Internet of Things Journal*, vol. 12, no. 22, pp. 47787-47803, 2025.
- Y. Zhang**, Z. Na, S. Li, B. Lin, Y. Lin, and A. Nallanathan. "Joint service caching and task offloading for multi-UAV-assisted offshore edge computing networks," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 12, pp. 19667-19680, 2025.
- Y. Zhang**, Z. Na, W. Lu, and C. Ji. "One friendly UAV against multiple mobile Eves in HAP networks: A novel physical layer secure communication scheme," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 1, pp. 1352-1364, 2025.
- Y. Zhang**, Z. Na, X. Du, Y. Lin, and B. Lin. "Secure task offloading for multi-layer aerial-offshore MEC networks with imperfect eavesdropper information," *IEEE Wireless Communications Letters*, vol. 14, no. 11, pp. 3814-3818, 2025.
- Y. Zhang**, Z. Na, Z. Wen, A. Nallanathan, and W. Lu. "Joint service caching, computation offloading and resource allocation for dual-layer aerial Internet of Things," *Computer Networks*, vol. 257, Art. no. 110974, 2025.
- Y. Zhang**, Z. Na, and Y. Wang. "Joint power allocation and deployment optimization for HAP-assisted NOMA-MEC system," *Wireless Networks*, vol. 30, no. 6, pp. 5007-5019, 2024.
- Y. Zhang**, Z. Na, H. Ren, X. Du, and L. Liu. "Computation offloading optimization for multi-layer aerial Internet of Things based on deep recurrent Q-network," *Radio Communications Technology*, pp. 1-9, 2025 (in Chinese).

Co-authored Journal Articles

1. J. Yin, Z. Na, and **Y. Zhang**. “Multi-UAV energy consumption minimization for multi-layer aerial wireless powered MEC: An online stochastic optimization approach,” *IEEE Internet of Things Journal*, vol. 13, no. 14, pp. 31707-31720, 2026.
2. Z. Wen, Z. Na, and **Y. Zhang**. “Energy efficiency optimization for RIS-assisted uplink-NOMA UAV network,” *IEEE Wireless Communications Letters*, vol. 13, no. 6, pp. 1640-1644, 2024.
3. X. Du, Z. Na, N. Zhang, and **Y. Zhang**. “Collaborative task offloading for LEO satellite Internet of Things: A novel computing coordinate graph-based approach,” *IEEE Systems Journal*, vol. 19, no. 3, pp. 848-859, 2025.

Conference Papers

1. **Y. Zhang**, Z. Na, H. Ren, and B. Lin. “Wireless-powered UAV-assisted MEC: Joint resource allocation and trajectory optimization,” *Proc. IEEE/CIC International Conference on Communications in China (ICCC)*, Dalian, China, pp. 1-6, 2023 (oral presentation).
2. J. Zhang, Z. Na, **Y. Zhang**, and S. Li. “Latency minimization for C-NOMA based multi-UAV MEC systems,” *Proc. IEEE 24th International Conference on Communication Technology (ICCT)*, Chengdu, China, pp. 1224-1228, 2024.
3. S. Li, Z. Na, **Y. Zhang**, and J. Zhang. “Latency minimization of multi-layer aerial networks for maritime collaborative mobile edge computing,” *Proc. IEEE 24th International Conference on Communication Technology (ICCT)*, Chengdu, China, pp. 202-206, 2024.

► MANUSCRIPTS UNDER REVIEW

1. **Y. Zhang**, Z. Wei, Z. Na, and B. Lin. “Digital-twin-assisted multi-timescale AI service deployment and inference orchestration for industrial edge networks,” *IEEE Transactions on Services Computing*, under review, Manuscript ID: TSC-2026-08-1174.
2. Z. Wei, **Y. Zhang***, R. He, and B. Lin. “Survivability-oriented task offloading and energy scheduling for tidal-powered island IoT,” *IEEE Communications Letters*, under review, Manuscript ID: CL2026-2529.
3. Z. Wei, **Y. Zhang***, R. He, and B. Lin. “Forecast-aided resilience scheduling for tidal-powered consumer IoT with energy-aware base-station edge cooperation,” *IEEE Transactions on Consumer Electronics*, under review, Manuscript ID: TCE-2026-07-3030.

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► RESEARCH PROJECTS

- **National Natural Science Foundation of China** (No. 61971081): High-Altitude Platform Base Station Deployment for Wide-Area Emergency Communication. Role: Research Participant
- **Natural Science Foundation of Liaoning Province** (No. 2023-MS-124): Efficient multi-domain UAV deployment optimization for 6G aerial networks. Role: Research Participant.

► HONORS AND ACADEMIC SERVICE

- **Honor:** Outstanding Graduate of Dalian Maritime University, 2025.
- **Professional memberships:** Member of the China Computer Federation and the Chinese Institute of Electronics.
- **Editorial service:** Part-time Academic Editor, Radio Communications Technology.
- **Peer review:** Reviewer for IEEE Internet of Things Journal and IEEE Transactions on Vehicular Technology.

► TECHNICAL SKILLS

- **Programming and Software:** Python, MATLAB, PyTorch, Gurobi, and CVX/CVXPY
- **Optimization and Learning Methods:** convex and mixed-integer optimization, generalized Benders decomposition, and deep reinforcement learning
- **System Modeling and Evaluation:** system-level simulation and statistical performance evaluation of wireless edge networks